```
'Ke'Shawn Stafford
'Project 2
'02/13/2024

' {$STAMP BS2}
' {$PBASIC 2.5}

G1          PIN 0
G2          PIN 1
Y1          PIN 2
Y2          PIN 3
R1          PIN 4
R2          PIN 5
PING        PIN 13
LCD         PIN 14
SERVO       PIN 15


Cm          CON 2260
distance    VAR Word
pulse       VAR Word
i           VAR Byte
note        VAR Byte

SEROUT LCD, 84, [22, 12]

DO
    PULSOUT PING, 5
    PULSIN PING, 1, pulse

    distance = Cm ** pulse

    SEROUT LCD, 84, [128, 1]
    SEROUT LCD, 84, [128, "Distance: ", DEC3 distance, " cm    "]

    note = 231 - (distance/12)

    IF distance >= 150 THEN
        GOSUB SolidG
    ELSEIF distance <= 149 AND distance > 100 THEN
        GOSUB FlashG
    ELSEIF distance <= 99 AND distance > 50 THEN
        GOSUB FlashY
    ELSEIF distance <= 49 AND distance > 20 THEN
        GOSUB FlashR
    ELSEIF distance <= 19 THEN
        GOSUB FlagR
    ENDIF

    PAUSE 50
LOOP

SolidG:
    HIGH G1
    HIGH G2
    LOW Y1
    LOW Y2
    LOW R1
    LOW R2
    SEROUT LCD, 84, [148, "Move UP      "]
    RETURN

FlashG:
    LOW Y1: LOW Y2
    FOR i = 1 TO 5
        HIGH G1
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        LOW G2
        SEROUT LCD, 84, [209, note]
        PAUSE distance +20
        LOW G1
        HIGH G2
        SEROUT LCD, 84, [209, note]
        PAUSE distance   +20
    NEXT
    PAUSE 5
    SEROUT LCD, 84, [148, "Move Slowly  "]
    RETURN

FlashY:
    LOW G1: LOW G2
    LOW R1: LOW R2
    FOR i = 1 TO 5
        HIGH Y1
        LOW Y2
        SEROUT LCD, 84, [209, note]
        PAUSE distance +20
        LOW Y1
        HIGH Y2
        PAUSE distance  +20
        SEROUT LCD, 84, [209, note]
    NEXT
    PAUSE 5
    SEROUT LCD, 84, [148, "CAUTION    "]
    RETURN

FlashR:
    LOW G1
    LOW G2
    LOW Y1
    LOW Y2
    FOR i = 1 TO 5
        HIGH R1
        LOW R2
        SEROUT LCD, 84, [209, note]
        PAUSE distance +20
        LOW R1
        HIGH R2
        PAUSE distance +20
        SEROUT LCD, 84, [209, note]
    NEXT
    PAUSE 5
    SEROUT LCD, 84, [148, "STOP NOW!   "]
    RETURN

FlagR:
    HIGH R1
    LOW R2
    FOR i = 1 TO 3
        PULSOUT SERVO, 800
        SEROUT LCD, 84, [209, note]

    NEXT
     PAUSE distance + 20
    LOW R1
    HIGH R2
    FOR i = 1 TO 3
        PULSOUT SERVO, 600
        SEROUT LCD, 84, [209, note]
    NEXT
    PAUSE distance + 20
    LOW G1
    LOW G2
```

```
LOW Y1
LOW Y2
SEROUT LCD, 84, [148, "STOP NOW!   "]
RETURN
```